

Predicting the Zero-Shot Transfer of a Promptable Video Tracker

Forecasting SAM 3's per-species, per-place reliability —
for detection and association — from label-free distances

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Abstract

Promptable trackers such as SAM3 segment and track any named animal in a video with no training on that species, yet no rule tells a practitioner *in advance* whether the model is reliable for a given animal in a given place. This dissertation asks whether such a rule exists. On the SA-FARI benchmark, whose splits are disjoint by species and location, we measure SAM3’s zero-shot promptable-tracking accuracy with the official evaluator, decompose it into detection (pDetA) and association (pAssA), and test whether each is predicted, *before running the model* and without any target label, by four distances (taxonomic, visual, environment, temporal). Under a deliberately powerful test — whole-species and whole-location hold-out with group-cluster-bootstrap significance — **it is not**: no distance beats a mean-predictor baseline out of sample, and the one apparent effect is an animal-size confound. A follow-up pivot to intrinsic scene difficulty (low-light, clutter) collapses to the same size effect, and the decomposition shows association carries almost no signal distinct from detection. The contribution is thus a rigorous, decomposed *negative* result — the first for a promptable video tracker — validated by a test shown to detect the one real signal (object size) when it is present.

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Chapter 1

Introduction

1.1 Motivation

Promptable video trackers — of which SAM3 [5] is the current state of the art — segment and track any named animal in a video with no training on that species, and conservation teams now deploy them on new species and regions constantly. Their appeal is exactly this zero-shot reach; their liability is that it is *unpredictable*, varying sharply from one animal or place to the next. Yet no rule tells a practitioner, *before running the model*, whether its output is trustworthy for a given animal in a given place — so a team pointing a tracker at a new species cannot tell in advance whether to trust it or whether the costly alternative, manual annotation, is still required. This dissertation studies that gap on the leading such tracker (SAM3), and then asks whether whatever we find is specific to it.

1.2 Problem statement and research question

This dissertation asks whether a promptable tracker’s zero-shot accuracy on an unseen (species, place) can be *predicted in advance* from properties measurable without any label of the target, and which factors govern that transfer — separately for *finding* the animal and *following* it. We study this concretely on SAM3 as the primary case, measuring novelty against the SAFARI reference split (Section 2.6), and then test whether the answer generalises beyond one dataset and one model — to two further datasets and, via a controlled model swap, to two other promptable trackers (Section 4.11).

1.3 Hypotheses

H0 (null)

the label-free distances carry no predictive information about pDetA/pAssA — transfer is not distance-driven.

H1 (detection)

pDetA falls with species novelty (taxonomic + visual distance).

H2 (association)

pAssA falls with environment difficulty (scene, day/night/IR, clutter, camera motion).

Failing to reject H0 is itself an informative result — and, as Chapter 4 shows, it is what we find: neither a novelty distance nor a follow-up *intrinsic-difficulty* pivot predicts transfer beyond a trivial animal-size effect. We even read familiarity from SAM3’s own features, and it too is null; a fuller *representational* probe — of what those features structurally encode — remains the open branch for future work.

1.4 Contributions

To our knowledge this is the first study to *predict*, before the model is run and from label-free distances, the zero-shot transfer of a promptable video tracker — and the first to split that prediction into **detection** (pDetA) and **association** (pAssA). This distinguishes it from label-free performance prediction for classifiers, which estimates a single scalar from the model’s own outputs on the target [2, 8, 14], and from the very recent static-image detector/segmenter estimators [20, 25]; and from camera-trap shift studies, which measure but do not forecast per-site reliability [3, 4, 10]. Concretely we deliver:

- (i) a rigorous, out-of-sample *negative* result — under a deliberately powerful test (whole-species and whole-location hold-out with group-cluster-bootstrap significance), label-free before-running distances do *not* predict SAM3’s transfer;
- (ii) the detection-versus-association decomposition, tested on two near-orthogonal splits — a constructed species hold-out (novelty) and the official location hold-out (environment);
- (iii) a *hardened* account of *why* it fails: a follow-up intrinsic-difficulty pivot collapses to the same animal-size confound, so the only label-free property that predicts detection is object size — the very nuisance the analysis controls for — and that positive control confirms the test detects signal when it exists;
- (iv) evidence that the null is not an artefact of *one* dataset or *one* model: it holds across two datasets (SA-FARI and BURST) and, in a controlled model swap on BURST, across two architecturally distinct promptable trackers (GLEE and Florence-2 + SAM2), so it is task-general rather than a quirk of SAM3.

1.5 Dissertation outline

Chapter 2 reviews the related work; Chapter 3 describes the data, measurement, distance features, and modelling; Chapter 4 reports the results and out-of-sample validation; Chapter 5 discusses them; Chapter 6 concludes.

Chapter 2

Background and Related Work

This chapter places the dissertation among four bodies of work: promptable segmentation and tracking (the model we study), the tracking metric and its detection/association decomposition (how we score it), label-free prediction of out-of-distribution performance (the statistical idea we extend), and distribution shift in camera-trap vision (the application and the closest empirical neighbours). It closes by stating precisely the gap this work fills.

2.1 Promptable segmentation and tracking

The *Segment Anything* line moved segmentation from a fixed label set to an interactive, promptable formulation, and its successors extended prompting from points and boxes to *concepts*. SAM3 [5] is a promptable model that, given a short text prompt such as "impala", detects, segments, and tracks every matching instance across a video with no training on that category — a *zero-shot*, open-vocabulary detect–segment–track capability. It is exactly this capability that conservation teams now deploy on new species and regions, and it is the (frozen) model whose transfer we predict. SAM3 was released alongside SA-FARI [21], a large multi-animal camera-trap tracking dataset that we use as the substrate for our natural experiment (Section 2.6). We treat SAM3 strictly as a fixed black box: we never fine-tune it, and the only quantity we fit is a small regression *about* its behaviour.

2.2 The tracking metric and its decomposition

Higher Order Tracking Accuracy (HOTA) [12] evaluates multi-object tracking by combining, over a range of localisation thresholds, a *detection* term (**DetA**: were the right objects found?) and an *association* term (**AssA**: were identities kept consistent over time?). SA-FARI scores promptable tracking with a promptable variant, **pHOTA**, that decomposes identically into **pDetA** and **pAssA**. This decomposition is the intellectual core of the dissertation: rather than predict a single scalar, we ask whether *finding* and *following* an animal transfer for *different reasons* — the former with how novel the animal is, the latter with how difficult the scene is — and we therefore model the two components separately throughout.

2.3 Predictable out-of-distribution generalisation

A now-mature literature shows that a model’s performance under distribution shift can be anticipated *without* target labels. Miller et al. [14] observe that out-of-distribution (OOD) accuracy is strongly linearly correlated with in-distribution accuracy across many models and shifts, and Baek et al. [2] show that the *agreement* between a pair of models tracks OOD accuracy closely enough to estimate it from unlabelled data alone. A parallel strand estimates accuracy from a

model’s own statistics on the unlabelled target: average thresholded confidence [8] calibrates a confidence threshold on source data and reads off target accuracy as the mass above it, while Deng and Zheng [7] regress classifier accuracy directly on a *distribution-shift feature* — the Fréchet distance between train and test feature sets — which is the closest methodological cousin to our own “regress the score on a distance” approach.

Two properties characterise essentially all of this work, and both are limitations we depart from. First, it targets **image classifiers** with a single scalar accuracy. The idea has only recently reached structured outputs: Yoo et al. [25] estimate object-detection performance without ground truth from the spatial consistency and confidence reliability of pre-NMS boxes, and reverse classification accuracy [20] predicts per-image segmentation quality by training a reverse model against a labelled reference. These are the closest structured-output predecessors, but both operate on *static single images*, neither concerns identity over time, and none decomposes the estimate into detection versus association. Second, every one of these methods still *runs the model on the target* and inspects its predictions, confidences, or agreement — it estimates the score of an *already-executed* model. We instead forecast from properties knowable *before the model is ever run* on the target.

2.4 Distribution shift in camera-trap and wildlife vision

That camera-trap models degrade at new locations is well established. *Terra Incognita* [3] showed that recognisers excellent at trained camera sites fail sharply at unseen ones; WILDS [11] consolidated this (via iWildCam) as a canonical real-world shift; and PanAf-FGBG [4] demonstrated that behaviour-correlated *backgrounds* causally drive the drop, proposing a latent-space mitigation. The claim that scene context, not the object, drives such failures has a longer pedigree in general vision. Xiao et al. [24] isolate foreground from background on ImageNet (the ImageNet-9 “Background Challenge”) and show that a classifier scores well above chance from the background *alone*, and that an adversarially chosen background flips a correctly classified foreground up to 87.5% of the time — background is genuine signal, not noise. Rosenfeld et al. [18] make the parallel point for detectors: transplanting an object into an unfamiliar scene suppresses its detection, flips its predicted identity with position, and disturbs *other* detections, exposing a reliance on context and co-occurrence. We adopt the same foreground/background decomposition these works rest on — masking the animal out to embed only the scene (the environment distance) and masking the scene out to embed only the animal (the visual distance) — but *invert its purpose*: where they perturb or subtract the background to *measure and mitigate* a model’s causal reliance on it [4, 18, 24], we use the isolated background as a label-free, before-running *distance* whose job is to *forecast* transfer. Notably, Xiao et al. [24] also find that more accurate, more robust models rely on background *less* — which anticipates our result for a strong, prompt-anchored zero-shot model, and to which we return in Chapter 5. These works are the empirical backdrop for our environment axis — but they share two features that mark the gap. They **measure or mitigate** degradation after the fact rather than *predicting* per-site or per-species reliability in advance, and they study **classifiers or behaviour recognisers**, not promptable trackers, scored by a single accuracy with no detection/association split. Most tellingly, a recent unified study of camera-trap recognition over time [10] explicitly names “predicting when zero-shot models will succeed at a new site” as an open question — and stops at naming it. This dissertation builds the predictor that question asks for.

2.5 What should predict transfer

Our candidate predictors are motivated by representation-learning results. Vision foundation models encode taxonomic structure across the tree of life [19], which motivates a *taxonomic* distance and, because appearance and phylogeny are correlated, a *visual* distance; we compute

the latter from mask-cropped DINOv2 [15] embeddings, with CLIP [16] as a robustness variant. The camera-trap location results above [3, 4] motivate an *environment* distance from the scene background, and per-video timestamps motivate a *temporal* gap. Crucially, all four are computable without any label of the target animal, and all are measured against a frozen reference set, so no target information can leak into a predictor.

2.6 The SA-FARI benchmark

SA-FARI [21] is a large open multi-animal camera-trap tracking dataset (11,609 densely annotated videos spanning 99 species with a seven-level taxonomy, 741 locations on four continents, and per-video timestamps), released with SAM3 and scored by the official VEval evaluator. It ships with abundant *hard negatives* — prompts for absent species that must return nothing — which we keep, as they drive the false-positive side of detection. Two properties of the shipped split shape our design and merit a precise statement. Inspection of the released annotations shows the train/test split is disjoint by **location** (test camera sites are unseen) but that its *species are largely shared* across the split; a genuine “unseen species” axis is therefore not directly available from the official split. Because SAM3 is frozen and zero-shot on *all* of SA-FARI, however, the split reflects nothing the model learned — it is only our choice of a *reference distribution* against which distances are measured. This licenses a species-disjoint *reference/probe partition* of the same videos for the primary (species-novelty) experiment, complemented by the official location split for the environment experiment; the two are near-orthogonal probes of the same data, and their construction is detailed in Chapter 3.

2.7 Summary and gap

Predictable OOD generalisation is established for classifiers scored by a scalar [2, 7, 8, 14] and, very recently, for static detectors and segmenters [20, 25]; camera-trap vision has thoroughly documented, but not forecast, location-driven degradation [3, 4, 10, 11]. No prior work predicts, *before running the model* and from label-free distances, the zero-shot transfer of a **promptable video tracker**, and none separates that prediction into **detection** and **association**. Asking that question rigorously — an out-of-sample test, over SA-FARI’s species-by-place-by-time structure, of whether label-free distances predict SAM3’s transfer, decomposed into pDetA and pAssA — is the contribution of this dissertation. As Chapter 4 reports, the honest answer is negative: transfer is not distance-driven.

Chapter 3

Data and Methodology

3.1 Research design and evaluation criteria

This work is a predictive-modelling study: we ask whether label-free *distances* measurable in advance (X) predict a frozen tracker’s zero-shot performance ($Y = \text{pDetA}/\text{pAssA}$), and which factors govern that transfer for *finding* versus *following* an animal. Because the target is a bounded score and the observations are grouped by species and location, the analysis is a weighted regression with group-aware, out-of-sample validation rather than a tuned tracker.

Definition of success and the decision rule. The intended deliverable is a *validated, out-of-sample* predictor of $\text{pDetA}/\text{pAssA}$ from label-free distances, with honest confidence intervals. A distance is taken to predict transfer only if it satisfies **both** criteria:

- (i) its coefficient’s group-cluster-bootstrap 95% confidence interval excludes zero; *and*
- (ii) the grouped cross-validated model beats a mean-predictor baseline by a margin that is significant under a paired group-bootstrap.

H1 (detection $\leftarrow$ species novelty) and **H2** (association $\leftarrow$ environment difficulty) are supported only when their respective distances meet this bar on the split that isolates them; if neither does under these deliberately powerful tests, we conclude **H0** — distance explains little variance — which is itself a valid result and motivates the representational-probing direction of Chapter 5.

3.2 Dataset, splits, and the analysis unit

We use SA-FARI [21] throughout. In its annotations, each test is a *probe* — a (video, text-prompt) pair; the prompt resolves the animal’s species and its seven-level taxonomy, while each video’s location and timestamp give place and time. We aggregate scores to a *cell* — a (species, location, time) group, keyed by a stable numeric species identifier — and attach support counts (n_{frames} , n_{masklets} , n_{videos}) used for weighting and as covariates. Hard negatives (prompts for absent species) are kept, as they drive the false-positive side of detection; the one exception is the stratified per-species cap used to make the large pooled pass feasible, under which hard negatives are set aside from inference and analysed instead by the separate false-positive pass (Section 3.7).

Two complementary splits. Inspection of the annotations shows the shipped split is disjoint by *location* but shares species across train and test (Section 2.6); on that split species-distance is ≈ 0 for every cell, so it cannot test species novelty. Because SAM3 is frozen and zero-shot on all of SA-FARI, the split only defines our *reference distribution*, so we are free to repartition the same videos. We therefore use two near-orthogonal experiments and run inference only once (scoring is per-video and split-agnostic):

Split A — species hold-out (primary).

Pooled species are partitioned by a *leave-species-out* view, so a held-out species’ distance is to the nearest *other* species; species-distance varies while the environment stays largely familiar, isolating H1. Validated by leave-species-out cross-validation.

Split B — location hold-out (secondary).

The official train/test split, whose unseen locations isolate H2 and whose benchmark comparability carries the initial score-sanity check.

Split A is a *constructed* hold-out — legitimate because the model is frozen, and reported as such. At the scale we work with, the location split (Split B) yields 1,447 scored cells over 113 taxonomy-bearing category identities (99 species) and 92 unseen test locations: 346 are *positive* cells — the queried species is present, so pDetA/pAssA are defined — and 1,101 are hard-negative cells, drawn from 1,486 hard-negative probes. The species hold-out (Split A) pools the present species of both official splits into 1,025 scored cells across the 99 species.

3.3 Measuring transfer

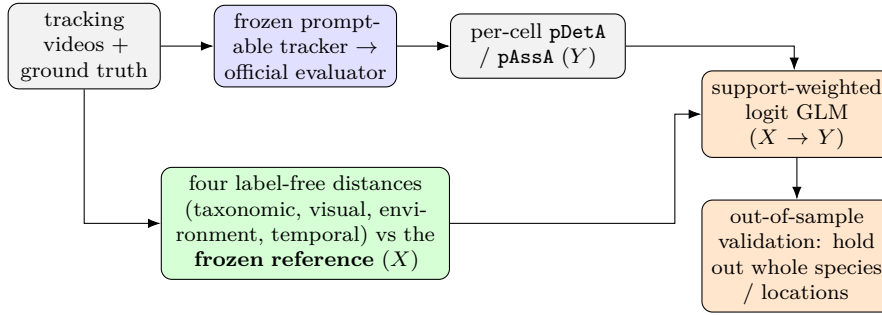


Figure 3.1: The measurement-and-modelling pipeline, identical for every dataset and tracker. *Top:* the frozen tracker is scored by the official evaluator into per-cell detection/association scores (the target Y). *Bottom:* four label-free distances are computed against a frozen reference (the predictors X). A support-weighted logit GLM relates X to Y , validated out of sample by holding out whole species and locations. Only the GLM is fitted; the tracker is never trained.

We now instantiate this pipeline on the primary case — SAM3 on SA-FARI — whose concrete choices, numbers, and benchmark status the rest of this section reports; the external replications and the model swap (Sections 3.8–3.8.1) then re-run the *same* pipeline on other datasets and trackers. SAM3 [5] is run frozen over the probe videos — species-specific prompts as the primary condition, a generic “animal” prompt for robustness — and scored with the official VEval evaluator, which we never re-implement. This yields a per-cell pDetA/pAssA/pHOTA [12] with its support. A single inference pass over the union of the Split A and Split B probes serves both experiments. Predictions are stored per (video, species) so that the several species-prompts issued on one video do not collide; the harness is resumable at per-video granularity and never updates a weight.

We aggregate the frame-level scores two ways and keep the two distinct. The *micro* score is the dataset-level VEval figure — the published SA-FARI operating point, ≈ 0.65 mask-pHOTA — which we report only to confirm we reproduce the benchmark. The *macro* score is the per-cell, support-weighted mean over positive cells, and it is our modelling target: the law is fit cell-by-cell, so the cell is the unit that must carry the score. Because the macro mean is a positives-only per-cell average it sits lower (≈ 0.53); every measurement number we quote is the macro value, labelled as such.

No retraining; validity of the frozen protocol. One point is worth stating plainly, because it is easily misread: **SAM3 is never retrained or fine-tuned in this work.** The only object

we fit is the small support-weighted logistic regression that maps the label-free distances X onto the observed scores Y ; the tracker is the public checkpoint, run once, its weights untouched. Reusing that one inference across both splits is therefore sound rather than a leak — because the model is frozen, a cell’s score cannot depend on which reference partition we later overlay to compute its distances.

This also settles the benchmark’s status. SA-FARI is a held-out, *zero-shot* benchmark for the released SAM3: the SAM3 report [5] does not list SA-FARI among its training sources, and the SA-FARI paper [21] reports the released checkpoint zero-shot (the ≈ 0.65 mask-pHOTA ballpark we reproduce in measurement) *alongside* separately fine-tuned rows whose gain we deliberately forgo. We can therefore verify non-membership at the *dataset* level; we cannot verify it frame-by-frame, since SAM3’s full pretraining corpus is undisclosed. We do not claim otherwise: every distance is measured against the SA-FARI *train reference*, not against SAM3’s true (unknown) pretraining distribution, and the familiarity proxy (Section 3.5.6) is the hedge for exactly this gap.

3.4 The frozen reference (leakage firewall)

For each split the reference (“seen”) species, locations, and taxonomy are frozen once; every distance is then computed against that reference only, so no probe information can enter a feature — verified by a unit test that swaps the reference and checks the distances move as expected. We assert the axis that is genuinely disjoint (location for Split B; the held-out species for Split A) and *report* any residual species overlap honestly rather than asserting it away.

3.5 Label-free distance features

All distances are label-free for the target and computed against the frozen reference.

3.5.1 Taxonomic distance

Lowest-common-ancestor tree steps from the probe’s species to the nearest reference species (shared genus = 1, family = 2, order = 3, ...); the leave-species-out reference keeps this non-trivial for shared species.

3.5.2 Visual distance

Each species is fingerprinted by a *prototype* — the re-normalised mean DINOv2 [15] embedding of its *ground-truth-mask-cropped* animals — and the visual distance is the cosine gap from a probe species’ prototype to that of the nearest *other* reference species (leave-species-out on Split A), with a CLIP [16] encoder as a robustness check. Using the dataset’s ground-truth masks means this feature needs no SAM3.

3.5.3 Environment distance

Embedding distance between the animal-masked-out background scene and the nearest reference location, plus interpretable scene covariates from the same frames: a day/night–infrared (IR) flag and a low-light fraction (both from per-channel colour statistics of the frame), and a clutter score given by the variance of a Laplacian over the masked background — a cheap measure of local edge/texture density, not a colour statistic.



Figure 3.2: The two image-based distances on a camera-trap frame (a collared peccary). (a) ground-truth mask (orange); (b) the mask-cropped animal the *visual* distance fingerprints; (c) the animal mean-inpainted, which the *environment* distance fingerprints. None of these features runs SAM 3.

3.5.4 Temporal gap

The minimum absolute year gap between a cell’s capture year (from the per-video creation-datetime metadata) and the nearest reference year. On the location split the timestamp is present for few cells and near-constant among them, so the temporal axis is reported there as *untested* rather than as a tested null.

3.5.5 Animal size (confound control)

Mean log ground-truth mask area per species. This is not a hypothesis variable but a nuisance covariate, included to separate genuine visual novelty from the fact that visually-distinctive species tend to be large and thus easier to segment; its role in exposing the one apparent effect as an artefact is Section 3.6.

3.5.6 SAM 3 familiarity proxy

Because SAM 3’s true pretraining corpus is undisclosed, every distance above is measured against the SA-FARI reference, not against what SAM 3 actually saw (Section 3.9 caveat). The familiarity proxy hedges this by reading familiarity *from the model itself*: how separable each species is in SAM 3’s *own* vision-encoder feature space. The same mask-cropped animals used for the visual distance are pushed through SAM 3’s image encoder, mean-pooling the patch-token features to a 1024-d vector, and each species is scored three ways — *silhouette* (own-cluster compactness vs the nearest other species), *nearest-prototype* (the visual distance recomputed with SAM 3’s encoder rather than DINOv2), and *Mahalanobis* typicality against the reference feature Gaussian. Like the other distances it is **before-running** (an image property, not a function of SAM 3’s tracking output, so it is non-circular) and **label-free**. Each metric is validated at the exact leave-species-out bar (Bonferroni over the three), size-controlled, and put in a head-to-head against the DINOv2 visual distance: the decisive question is whether it adds any out-of-sample gain over the visual distance + size (Section 4.8).

3.6 Statistical modelling and hypothesis testing

3.6.1 Per-target regression

Each bounded target (pDetA, pAssA) is modelled *separately* with a support-weighted logit-link (fractional-logit) regression over the standardised distances plus the scene covariates and a $\log(n_{\text{frames}})$ covariate, so that “rare” is never mistaken for “far”. Detection and association are analysed apart because their decomposition is the intellectual core of the project. In practice association is a *near-degenerate* target on this data — most cells contain a single object,

so **pAssA** tracks **pDetA** almost exactly — and it is therefore reported as such rather than as an independently powered test (Chapter 5).

3.6.2 Honest, group-aware inference

Naive model confidence intervals are invalid here for two reasons: support-weighting by frame count inflates the effective sample size, and the distances are constant within a species or a location, so the hundreds of cells are really only tens of independent groups (pseudo-replication). We therefore report *group-cluster-bootstrap* intervals — resampling whole species and whole locations, refitting, and taking percentile intervals — as the honest confidence interval for every coefficient.

3.6.3 Out-of-sample validation and the significance test

The fitted models are validated *out of sample* by holding out whole groups (leave-species-out on Split A, leave-location-out on Split B) and predicting the held-out group. The out-of-sample gain over a mean-predictor baseline is tested with a *paired group-bootstrap* that resamples whole held-out groups, yielding a confidence interval and a one-sided *p*-value on the gain — the criterion (ii) of Section 3.1.

3.6.4 Variance attribution and confound control

Variance is attributed to each factor by dominance/commonality analysis with a VIF collinearity check — not by four univariate fits — and raw correlations are reported as a model-free cross-check. Where a distance reaches significance, we refit with the animal-size covariate to test whether the effect is a size confound rather than novelty.

3.6.5 The intrinsic-difficulty pivot and the nested decomposition

Because the sole apparent effect proved to be a size confound, we ask a complementary question: is transfer governed by the intrinsic *difficulty* of the target imagery rather than its *novelty*? We promote the label-free scene properties already measured for the environment axis — the low-light fraction, the clutter score, and the night/IR flag — from nuisance covariates to predictors of interest, and test them against the same out-of-sample bar. To separate a genuine difficulty effect from the size confound, we fit a *nested* sequence of models — support only; size only; difficulty only (no size); difficulty plus size — and compare their out-of-sample gains, so that any apparent difficulty effect that is really size is exposed as such (Chapter 4, Table 4.4). We also report each scene property’s *cell-level* correlation alongside its location-level value, because aggregation inflates correlations (the ecological-correlation fallacy).

Detecting a signal, not asserting its absence. These tests are demonstrably able to detect a real effect: the animal-size covariate not only has an interval excluding zero but *validates out of sample* at the same whole-group hold-out bar the distances fail (Chapter 4), and the estimation and cross-validation machinery is unit-tested to recover planted effects on synthetic data. A null under a test that fires for size is therefore evidence of *absence of signal*, not of *absence of power*.

3.6.6 Design-robustness checks

To show the null is not an artefact of a single design choice, we re-run the entire hold-out procedure with one component of the pipeline swapped at a time: the visual encoder (DI-NOv2 → CLIP), the crop (mask-cropped animal → a whole-frame bounding box that keeps the background), the prompt (species-specific → a generic “animal”), and the visual-distance functional itself (nearest-prototype cosine → a distributional Fréchet/MMD distance between the

target and reference embedding sets, after AutoEval [7]). Each variant is fitted and validated at the identical whole-group hold-out bar of Section 3.1, with significance judged after a Bonferroni correction over the variant family; the outcomes are reported in Chapter 4 (Table 4.11).

3.7 Detection precision: false positives on hard negatives

The **pDetA** model conditions on the animal being present, so it measures recall-given-present, not hallucination. On hard negatives (the queried species is absent) a correct run returns no masklet; we read the prediction directly, flag any returned masklet above a score threshold as a false positive, attach each species’ taxonomic and visual distance, and test — at the species level, to avoid pseudo-replication — whether the hallucination rate rises with novelty.

3.8 Independent replication on external datasets

Every SA-FARI result reuses one frozen inference pass on one dataset, so the two splits are near-orthogonal but not statistically independent. To test whether the before-running null is a property of the transfer problem rather than of SA-FARI, we re-run the *entire* pipeline — unchanged frozen SAM 3, the same official scorer, the same distances, and the same leave-species-out cross-validation and paired group bootstrap — on an independent tracking dataset. The abstraction boundary that makes this cheap is the annotation schema, not the loader: an offline adapter converts the dataset into the SA-Co annotation JSON (the SA-FARI/VEval schema) the loader and evaluator already read, so nothing downstream changes. Only the small logit GLM is ever fitted; SAM 3 is frozen throughout.

BURST (mask-native, many-species). The animal subset of the BURST validation split [1] (built on TAO [6]): 41 animal species across 132 video $\times$ class cells, with native COCO-RLE masks scored by mask-HOTA. The animal categories are selected from BURST’s 482 LVIS classes [9] by a WordNet filter [13] (hyponyms of *animal.n.01*) and mapped to a seven-level biological taxonomy where resolvable (25/41 species full, the rest left undefined). BURST carries no site or timestamp metadata, so the cell key is the per-video sequence and leave-species-out is the only valid grouping; the association target is genuinely non-degenerate here, so both components are tested. Masks stream from the RWTH server and TAO frames from the Hugging Face mirror by HTTP range, so only the annotated frames of the capped animal videos are fetched.

Interpreting the replication. BURST is held to the exact same out-of-sample bar as SA-FARI, with the animal-size confound as the positive control — it validates out-of-sample on SA-FARI, so a test that stays flat for the distances there is genuinely finding no signal; where it too fails to fire (as on BURST) the design is simply too small to certify power, marking the result as consistency rather than confirmation. We report convergence, not independent proof: BURST is better powered by cell count than the SA-FARI location split but still cannot exclude a small before-running distance effect (Section 4.9).

3.8.1 The controlled model swap

The dataset-level analysis varies the data while holding the tracker fixed; to test whether the null is a property of SAM3 specifically we also vary the *tracker*. A *controlled model swap* holds everything else constant — the same cells, the same ground truth, the same four label-free distances, and the same support-weighted GLM under leave-species-out cross-validation — and substitutes SAM3 with a second open-vocabulary, text-promptable tracker; only the target **pDetA** changes, since the distances are functions of the data and ground truth, not of the tracker. This is a controlled swap, not an independent replication: it reuses the same cells and

ground truth, so it isolates the tracker as the sole varying factor. We swap in two challengers of deliberately different architecture — GLEE [22], a single unified detect–segment–track model, and Florence-2 [23] + SAM 2 [17], a caption-style box generator with no confidence score followed by a mask-propagating tracker.

Two safeguards keep the swap fair and honest. First, a *measurement gate*: before fitting any regression on a challenger’s scores we require its per-cell `pDetA` to be non-degenerate (real spread across cells, not floored at zero); a regression on degenerate scores has no variance to explain and would report a spurious “null,” so a challenger that scores ≈ 0 on a dataset is dropped from the swap on that dataset rather than modelled. Second, *scoring fairness*: because open-vocabulary detectors’ confidence scores are calibrated to their own training domain, we verify that any near-zero challenger score is a genuine detection failure and not an artefact of the evaluator’s fixed confidence gate — by re-scoring under a box-based metric and under a threshold-free average-precision metric with no gate at all. This gate is why the swap runs on BURST, not SA-FARI: SA-FARI is SAM3’s own co-released benchmark and its camera-trap domain collapses the challengers’ calibration even where they localise well, so it cannot yield a fair, non-degenerate challenger score, whereas SAM3 scores no lower on the independent BURST than on SA-FARI. Every challenger configuration is fixed in advance from the model’s own defaults, run once, and reported as-is (Section 4.11).

3.9 Reproducibility, implementation, and constraints

The model is frozen and scored by the official evaluator; the reference (seen) set is frozen before any distance is computed; hard negatives are retained; and all inference is single-GPU and inference-only. The pipeline is config-driven (every constraint is a toggle, so an ablation flips config, not code) and covered by hermetic tests; every reported number is regenerated from the repository, and split definitions, seeds, and reference sets are persisted so both experiments are reproducible end to end. The engineering safeguards that keep the measurement honest — memory-lean feature assembly, per-(video, species) prediction keying, trimming the ground truth to the scored videos before evaluation, and the size-confound control — are catalogued in the appendix, each framed as a validity safeguard.

Chapter 4

Results and Evaluation

4.1 Measurement

As an initial score-sanity check, zero-shot SAM3 on the SA-FARI test split reproduces the published SAM3 ballpark; the support-weighted per-cell aggregate is in Table 4.1.

Table 4.1: Zero-shot SAM3 on the SA-FARI test split — support-weighted mean over 346 scored cells (of 1447).

Metric (mask)	Value
pDetA	0.527
pAssA	0.546
pHOTA	0.533

4.2 The fitted law

Table 4.2 gives the standardised logit-GLM coefficients for detection and association.

Table 4.2: Standardised logit-GLM coefficients (group-cluster-bootstrap 95% CIs) for detection and association. Every distance’s CI spans zero.

Factor	pDetA coef. [CI]	pAssA coef. [CI]
Taxonomic distance	-0.362 [-3.87, 0.09]	-0.368 [-3.40, 0.08]
Visual distance	-0.144 [-0.50, 0.14]	-0.122 [-0.49, 0.18]
Environment distance	0.042 [-0.19, 0.24]	0.013 [-0.23, 0.21]
Clutter	-0.066 [-0.38, 0.29]	-0.087 [-0.41, 0.28]
Night / IR	-0.405 [-0.94, 0.11]	-0.462 [-1.02, 0.11]
log(support)	-0.216 [-0.51, 0.17]	-0.224 [-0.53, 0.18]

4.3 Variance partitioning and the intrinsic-difficulty decomposition

A dominance (LMG/Shapley) analysis confirms the distances carry almost nothing: together the four explain only $\approx 3\%$ of the variance in detection, no single factor dominates, and every VIF is ≈ 1 (Table 4.3) — so the null is not an artefact of collinearity masking a real effect. Because the one apparent effect (visual distance, below) turned out to be an animal-size confound, we pivot

Table 4.3: Variance partition for detection: each distance’s unique share of R^2 (LMG/Shapley) and its VIF. The four distances explain only $\approx 3\%$ of the variance and every VIF ≈ 1 — the null is not masked by collinearity.

Factor	unique R^2	share	VIF
Taxonomic	0.014	48%	1.06
Visual	0.012	39%	1.06
Environment	0.004	13%	1.00
Temporal	0.000	0%	—

the hypothesis from *novelty* to intrinsic *difficulty*: we promote the label-free scene properties — low-light/IR (`achromatic_fraction`), clutter, and the night/IR flag, each computed per location from the frames with no SAM3 run and no target label — from nuisance covariates to predictors of interest, and test them at the same bar. Table 4.4 decomposes the outcome. Taken *with* an animal-size covariate each difficulty signal *appears* to validate (leave-species-out and leave-location-out $\Delta\text{MAE} \approx +0.017$, both significant), but a nested decomposition shows the gain is entirely the size covariate: log-area *alone* carries it ($\Delta\text{MAE} = +0.017$, $p = 0.001/0.014$), while low-light *alone* does *not* validate ($p = 0.12/0.23$) and adds only $\approx +0.002$ over size. The low-light effect is also far weaker than it first looks: the location-level night/IR–pDetA correlation of $r = -0.377$ is an aggregation artefact (the ecological-correlation fallacy) — at the cell level it is only -0.13 , and clutter’s cell-level correlation is 0.00 . The intrinsic-difficulty pivot is therefore **also null**: the only label-free property that predicts detection is animal size.

Table 4.4: Nested size decomposition of the intrinsic-difficulty test (detection, 346 cells). Entries are the out-of-sample MAE gain over a mean baseline, Δ [95% CI] (p), under leave-species-out and leave-location-out. The entire gain sits in the size covariate; low-light *alone* does not validate — the difficulty signals pass only by riding on size.

Model	Leave-species-out Δ [CI] (p)	Leave-location-out Δ [CI] (p)
log(support) only	+0.001 [-0.007, +0.008] (0.432)	+0.002 [-0.005, +0.008] (0.291)
log(area) only [<i>size</i>]	+0.015 [+0.002, +0.028] (0.014)	+0.017 [+0.009, +0.025] (0.001)
Low-light only (no size)	+0.003 [-0.006, +0.011] (0.233)	+0.004 [-0.003, +0.011] (0.123)
Low-light + size	+0.017 [+0.004, +0.032] (0.011)	+0.020 [+0.011, +0.030] (0.000)

4.4 Out-of-sample validation

Table 4.5 reports whole-group hold-out error against a mean-predictor baseline. This is the headline result and it is a **null**: on both the location and the species schemes, and for both targets, no distance model beats simply predicting the mean — every ΔMAE is within noise of zero and no gain is significant under the paired group-bootstrap. **This is not an underpowered test.** The same GLM and grouped cross-validation, on the same cells, *do* detect a real signal when one exists: animal size (log-area) validates out of sample at the identical bar ($\Delta\text{MAE} = +0.017$, $p < 0.017$ on both schemes; Table 4.4). A test that fires for size yet stays flat for the distances is evidence of a genuine *absence* of before-running label-free signal, not of low power.

Table 4.5: Out-of-sample error: whole-group hold-out (leave-species-out / leave-location-out) vs a mean predictor. Δ MAE is the gain over baseline, with a paired-bootstrap 95% CI and one-sided p . No hold-out’s gain is significant.

Hold-out	Target	n	MAE	Baseline	Δ [95% CI]	p
location	pAssA	346	0.289	0.295	+0.007 [-0.005, +0.018]	0.13
location	pDetA	346	0.293	0.298	+0.004 [-0.007, +0.015]	0.23
species	pAssA	346	0.289	0.295	+0.006 [-0.006, +0.017]	0.16
species	pDetA	346	0.293	0.298	+0.004 [-0.006, +0.015]	0.24

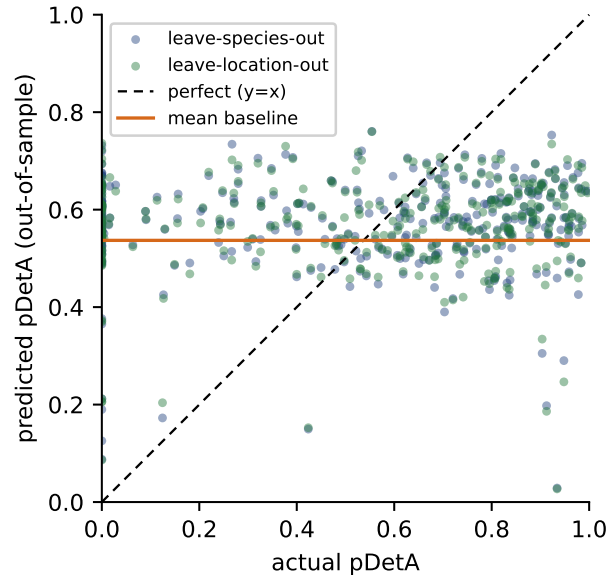


Figure 4.1: Out-of-sample detection predictions vs the truth, held-out species and locations. They collapse onto the mean baseline (orange) instead of the diagonal (dashed) — the distances carry no predictive power.

4.5 Species hold-out (H1): novelty and the size confound

The primary novelty test — leave-one-species-out over the pooled train+test species (99 species, 1025 scored cells) — is in Tables 4.6–4.8. Out-of-sample prediction is not significant on any hold-out (Table 4.8). The only coefficient whose CI excludes zero, *visual distance* (Table 4.7), is positive — meaning more-novel species are detected *better*, the reverse of what H1 predicts — and is a **confound**: visually-distinctive species are larger (visual $\leftrightarrow$ log-area $r = 0.22$) and larger animals score higher (log-area $\leftrightarrow$ pDetA $r = 0.30$, the strongest raw correlate of the score). Adding an animal-size covariate shrinks the visual coefficient from +0.37 to +0.22 (CI now spanning zero) while size itself is the significant driver, so “visual novelty helps” was really “bigger animals are easier”. Taxonomic novelty is flat ($r \approx 0.00$). Novelty does not predict transfer.

Table 4.6: Zero-shot SAM3 on the pooled train+test set — support-weighted mean over 1025 scored cells (of 2126).

Metric (mask)	Value
pDetA	0.536
pAssA	0.597
pHOTA	0.559

Table 4.7: Standardised logit-GLM coefficients (group-cluster-bootstrap 95% CIs) for detection and association, species hold-out.

Factor	pDetA coef. [CI]	pAssA coef. [CI]
Taxonomic distance	0.202 [-0.09, 0.43]	0.209 [-0.06, 0.42]
Visual distance	0.369 [0.11, 0.67]	0.271 [0.04, 0.54]
Environment distance	0.043 [-0.15, 0.17]	0.023 [-0.15, 0.13]
Clutter	-0.284 [-0.52, 0.02]	-0.230 [-0.46, 0.06]
Night / IR	-0.286 [-0.77, 0.22]	-0.439 [-0.92, 0.06]
log(support)	-0.020 [-0.14, 0.16]	0.051 [-0.06, 0.23]

Table 4.8: Out-of-sample error, species hold-out: whole-group hold-out (leave-species-out / leave-location-out) vs a mean predictor. Δ MAE is the gain over baseline, with a paired-bootstrap 95% CI and one-sided p . No hold-out’s gain is significant.

Hold-out	Target	n	MAE	Baseline	Δ [95% CI]	p
location	pAssA	1025	0.270	0.270	-0.000 [-0.006, +0.005]	0.52
location	pDetA	1025	0.274	0.274	-0.001 [-0.007, +0.006]	0.58
species	pAssA	1025	0.273	0.270	-0.003 [-0.014, +0.008]	0.74
species	pDetA	1025	0.277	0.274	-0.004 [-0.015, +0.009]	0.74

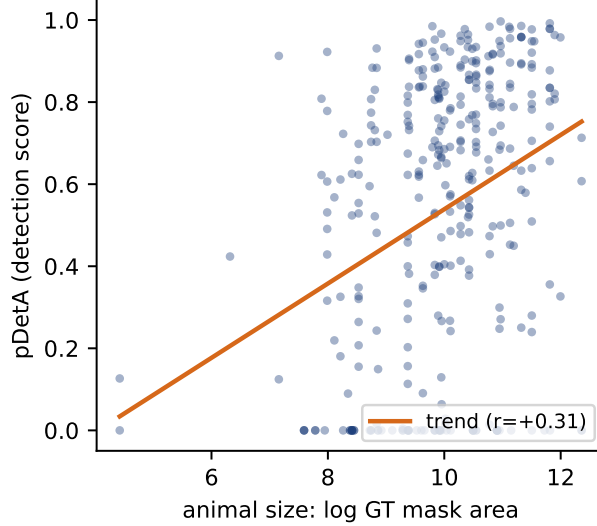


Figure 4.2: The one label-free property that predicts detection: animal size. Per-cell **pDetA** rises with log mask area ($r = +0.31$) — the confound behind the apparent visual-distance effect.

4.6 False positives / hallucination

The detection score conditions on the animal being present, so it measures recall, not hallucination. The 1486 *hard negatives* (queries for absent species) ask the complementary question, and are the closest the study comes to a validated before-running signal. SAM3 returns a masklet anyway about **9.8%** of the time. This hallucination rate does *not* rise with taxonomic novelty ($r \approx +0.06$, n.s.); it *falls* with visual distinctiveness ($r \approx -0.33$), and this survives an animal-size control (partial -0.37) — a genuine, size-independent, label-free correlate, though its direction (“distinctive = safer”) is the *opposite* of “novelty hurts”. Crucially it is a **correlate, not a predictor**: predicting a held-out species’ hallucination rate from visual distance beats a mean baseline by only $+0.008$ MAE and *just misses* out-of-sample significance ($p = 0.053$; Table 4.9). Even the study’s best signal does not clear the bar.

Table 4.9: False positives on the 1486 hard negatives (113 species): overall hallucination rate 9.8%. More visually distinctive species hallucinate *less* — a real, size-independent correlate — but it does not validate out of sample (leave-species-out $p = 0.053$): a correlate, not a predictor.

Test (species-level)	statistic	95% CI	verdict
FP rate vs taxonomic distance	+0.06	[−0.17, +0.23]	n.s.
FP rate vs visual distance	−0.33	[−0.43, −0.23]	significant
visual, size-controlled (partial)	−0.37	[−0.47, −0.28]	significant
FP rate vs animal size	+0.13	[−0.06, +0.31]	n.s.
leave-species-out OOS gain	+0.008	[−0.002, +0.017]	$p = 0.053$

4.7 Ablations and robustness

Factor ablation. Refitting detection with each distance dropped in turn barely moves the fit or the (null) out-of-sample gain (Table 4.10): no single distance carries signal. The only design that changes anything is adding the animal-size covariate, which alone lifts the out-of-sample gain to significance ($\Delta\text{MAE} \approx +0.02$, $p \leq 0.02$) — again consistent with size being the sole real correlate. Removing the support covariate does not rescue a distance effect either. The variance

partition (Table 4.3) tells the same story from the other direction.

Table 4.10: Factor ablation for detection: McFadden pseudo- R^2 and the out-of-sample MAE gain over a mean baseline under leave-species-out (LSO) and leave-location-out (LLO). Dropping any distance barely changes the fit; only the animal-size covariate moves the score.

Design	pseudo- R^2	LSO Δ (p)	LLO Δ (p)
full (4 distances)	0.055	+0.004 ($p=0.24$)	+0.004 ($p=0.23$)
drop taxonomic	0.039	+0.000 ($p=0.47$)	+0.000 ($p=0.49$)
drop temporal	0.055	+0.004 ($p=0.24$)	+0.004 ($p=0.23$)
drop visual	0.048	+0.004 ($p=0.22$)	+0.004 ($p=0.28$)
drop environment	0.055	+0.006 ($p=0.15$)	+0.005 ($p=0.17$)
+ animal size	0.119	+0.018 ($p=0.02$)	+0.020 ($p=0.00$)
– support covariate	0.035	−0.002 ($p=0.72$)	−0.003 ($p=0.73$)

Design robustness. The null is not an artefact of the encoder, the crop, the prompt, or the distance functional (Table 4.11). Re-running the full leave-species-out and leave-location-out procedure with a CLIP rather than DINOv2 encoder, a whole-frame rather than mask-cropped embedding, and a distributional Fréchet or MMD rather than nearest-prototype distance each leaves the out-of-sample gain indistinguishable from the baseline (all $\Delta\text{MAE} \leq 0.007$, $p \geq 0.10$, every 95% CI spanning zero). A generic “animal” prompt is included for completeness: prompting non-specifically collapses per-species detection to a near-constant, mostly-zero target (mean $\text{pDetA} \approx 0.14$, 73% of cells exactly zero),¹ so the distance model cannot beat — and, fitting a degenerate target, underperforms — a mean predictor. Across every variant the label-free distances fail identically, confirming the null is a property of the transfer problem, not of one modelling choice.

¹A minority of clips were additionally skipped for GPU memory on the 16 GB card used for this pass; they are scored as misses and add noise but do not drive the collapse, which is intrinsic to the non-specific prompt.

Table 4.11: Design-robustness sweep (detection). Each row re-runs the primary model with one design choice swapped; entries are the out-of-sample MAE gain, Δ [95% CI] (p), under leave-species-out and leave-location-out (Bonferroni $m = 5$). The null holds throughout — no encoder, crop, prompt, or distributional distance beats the mean.

Variant	Leave-species-out Δ [CI] (p)		Leave-location-out Δ [CI] (p)		Beats mean?
baseline (DINOv2, mask-crop, species prompt) (<i>baseline</i>)	+0.004	[-0.006, +0.015] (0.237)	+0.004	[-0.007, +0.015] (0.235)	n.s.
CLIP encoder	+0.004	[-0.006, +0.014] (0.247)	+0.004	[-0.007, +0.014] (0.270)	n.s.
whole-frame (bbox + background)	+0.002	[-0.008, +0.013] (0.349)	+0.002	[-0.008, +0.013] (0.357)	n.s.
generic “animal” prompt	-0.346	[-0.381, -0.309] (1.000)	-0.343	[-0.363, -0.326] (1.000)	n.s.
Frechet distributional distance	+0.007	[-0.005, +0.019] (0.120)	+0.007	[-0.004, +0.019] (0.102)	n.s.
MMD distributional distance	+0.002	[-0.008, +0.012] (0.374)	+0.002	[-0.009, +0.012] (0.349)	n.s.

4.8 SAM 3 familiarity proxy

The distances measure novelty against the SA-FARI reference, not against SAM 3’s undisclosed pretraining set. The familiarity proxy (§3.5.6) closes that gap by reading familiarity from SAM 3’s own vision-encoder feature space — three separability metrics on the 346 positive cells (Table 4.12). The result is another null: on the leave-species-out novelty test none of the three clears the Bonferroni threshold ($\alpha = 0.017$) — the strongest, silhouette, reaches only $p = 0.034$ ($\Delta\text{MAE} = +0.012$), and nearest-prototype and Mahalanobis do not come close ($p = 0.068, 0.085$). All three are dominated by the DINOv2 visual distance they re-encode ($r = 0.63\text{--}0.85$), and in the head-to-head none adds out-of-sample gain over `visual_distance` + size. The one curiosity is that silhouette, unlike the visual distance, is *not* a size artefact ($r_{\text{size}} = -0.16$) — it is the size-independent half of visual distinctness — yet it still fails to predict transfer.

The metrics do reach significance under leave-*location*-out ($p \leq 0.01$), but this is not a novelty test: the familiarity value is species-constant (within-species variance = 0), and on the location split the held-out locations’ species are shared with the training locations, so a species-constant feature is never tested on an unseen species. Crucially, on these same 346 cells the before-running animal-size control does validate out-of-sample, so the leave-species-out familiarity null reflects a genuine absence of signal, not a lack of power. The null replicates on BURST (all three metrics $p > 0.13$ except a matching silhouette near-miss at $p = 0.020$; none adds over visual + size). Reading familiarity directly from SAM 3’s representations therefore does *not* rescue before-running prediction — it recovers the same size-entangled visual-distinctness signal that was already H0, and closes the “but you do not know SAM 3’s true training set” loophole in the null.

Table 4.12: SAM 3 familiarity proxy (detection): does a species’ separability in SAM 3’s *own* feature space predict pDetA? Each metric is size-controlled; ΔMAE (p) is the out-of-sample gain over a mean predictor. None clears Bonferroni ($\alpha = 0.017$) on either dataset, and all three re-encode the visual distance ($r_{\text{vis}} = 0.63\text{--}0.85$). As the size control validates on the same cells, this is a genuine null, not an underpowered one.

Metric	SA-FARI ΔMAE (p)	BURST ΔMAE (p)	r_{vis}	r_{size}	clears bar?
silhouette	+0.012 (0.034)	+0.012 (0.020)	+0.67	−0.16	no
nearest-prototype	+0.009 (0.068)	+0.006 (0.136)	+0.63	+0.03	no
mahalanobis	+0.008 (0.085)	+0.006 (0.148)	+0.85	+0.32	no

4.9 Independent replication (BURST)

Every result so far reuses one frozen inference pass on one dataset, so the species and location experiments are near-orthogonal overlays rather than statistically independent replications. To test the null on genuinely independent data we re-ran the whole pipeline on **BURST** [1]. Its animal subset — the animal LVIS classes of the validation split, streamed mask-native and prompted by class name — yields 132 video $\times$ class cells across 41 species, in a distinct regime (internet/driving/movie video, not camera traps). Zero-shot SAM 3 is again non-degenerate (pHOTA 0.635, Table 4.13), and here the association target is genuine ($\text{corr}(\text{pDetA}, \text{pAssA}) = 0.76$; of the 71 multi-object cells, none has pAssA exactly equal to pDetA), so both components are testable.

A second convergent null. Under leakage-free leave-species-out cross-validation the before-running distances again fail to predict detection transfer: the combined distance model improves out-of-sample MAE by only +0.002 ($p = 0.374$), with visual (+0.003, $p = 0.257$), taxonomic (full-taxonomy subset −0.006, $p = 0.952$) and environment (−0.001, $p = 0.747$) all individually

null, and the association specs null likewise (Table 4.14). The result is robust to dropping the 19 single-cell species and to a size control. The lone near-signal, visual distance, is the familiar artefact: its correlation with detection is +0.17 — in the *opposite* direction to the novelty hypothesis (more-novel animals detected better), and a size confound (partial correlation given log-area +0.08, $p = 0.35$). Both datasets — SA-FARI and BURST — now converge on the same before-running distance null.

Table 4.13: Zero-shot SAM3 on the BURST val animal subset — support-weighted mean over 132 probe cells (41 species), mask-HOTA. The score is non-degenerate and the association target is genuine ($\text{corr}(\text{pDetA}, \text{pAssA}) = 0.76$, 71/132 multi-object cells).

Metric (mask-HOTA)	Value
pDetA	0.607
pAssA	0.688
pHOTA	0.635

Table 4.14: BURST independent replication — out-of-sample error under leave-species-out cross-validation (41 species, 132 mask-native cells) vs a mean predictor. ΔMAE is the pDetA gain over baseline, with a paired-bootstrap 95% CI and one-sided p . Every distance is null; not even the animal-size control reaches significance, so BURST is a convergent null that cannot certify power for a small species-level effect.

Predictor	ΔMAE	95% CI	p
distances (taxonomic + visual + environment)	+0.002	[−0.007, +0.011]	0.374
visual only	+0.003	[−0.005, +0.013]	0.257
taxonomic only (full-taxonomy subset, $n=82$)	−0.006	—	0.952
environment only	−0.001	[−0.005, +0.003]	0.747
animal size (log_area) [control]	+0.005	[−0.003, +0.014]	0.143

Convergent, but not proof. At 132 cells BURST supplies a mask-native regime with a genuinely non-degenerate association target, and the distance null holds throughout. It is not, however, positive proof: no before-running quantity — not even the animal-size control that validates on SA-FARI — reaches significance here, so we cannot independently certify that this test could have caught a small species-level effect. The lone near-signal, visual distance, is again the wrong-direction size confound (its correlation with detection is +0.37 between species, the opposite sign to novelty). We therefore read BURST as a second convergent null — no moderate-or-large before-running distance signal, in an independent mask-native regime with a non-degenerate association target — rather than positive proof, and note that a small species-constant effect cannot be excluded.

4.10 Cross-dataset synthesis

Table 4.15 collects every feature tested, its numerical out-of-sample result, and the dataset it was measured on. The pattern is consistent across both: no before-running label-free distance — taxonomic, visual, or environment, alone or combined — predicts detection or association transfer out of sample; every gain over a mean predictor is within bootstrap noise of zero. The only near-signal, visual distance, appears in the *opposite* direction to the novelty hypothesis (more-novel animals detected better, not worse) and dissolves under an animal-size control wherever it is examined. The only before-running quantity that ever reaches significance is the animal-size confound itself — and only modestly, on the large SA-FARI sample — which doubles as

the positive control that keeps the null honest: the same test that stays flat for the distances does pick up size out-of-sample. The contribution is this robust, twice-replicated null, honestly bounded: consistent across a camera-trap and an internet-video dataset, and powered enough on SA-FARI to exclude a moderate effect, but not a small one.

Table 4.15: What was tested, its out-of-sample result, and the dataset. Result is the leave-species-out Δ MAE over a mean predictor (one-sided bootstrap p) unless noted; positive-and-significant means the feature predicts transfer. The label-free distances are null on both datasets; only the animal-size confound reaches significance, modestly, on SA-FARI (the positive control). “Opposite direction” means more-novel animals are detected *better*, not worse — so it cannot support the hypothesis.

Feature tested	Result (leave-species-out)	Dataset
<i>Before-running label-free distances (the research question)</i>		
Combined distances (tax. + vis. + env.)	Δ MAE -0.004 ($p=0.74$), n.s.	SA-FARI (1025 cells, 99 sp.)
Combined distances (tax. + vis. + env.)	Δ MAE $+0.002$ ($p=0.37$), n.s.	BURST (132 cells, 41 sp.)
Visual distance alone	coef. -0.14 $[-0.50, 0.14]$, n.s.	SA-FARI
Visual distance alone	Δ MAE $+0.003$ ($p=0.26$) — opposite direction	BURST
Taxonomic distance alone	coef. -0.36 $[-3.87, 0.09]$, n.s.	SA-FARI
Taxonomic distance alone (full-tax. $n=82$)	Δ MAE -0.006 ($p=0.95$), n.s.	BURST
Environment distance alone	coef. $+0.04$ $[-0.19, 0.24]$, n.s.	SA-FARI
Environment distance alone	Δ MAE -0.001 ($p=0.75$), n.s.	BURST
Temporal gap	n.s. (present); degenerate elsewhere	SA-FARI (BURST: all-NaN)
<i>Confound control</i>		
Animal size (log-area)	Δ MAE $+0.015$ ($p=0.01$) — validates	SA-FARI
Animal size (log-area)	Δ MAE $+0.005$ ($p=0.14$), n.s.	BURST
<i>Detection precision (hard negatives)</i>		
Hallucination rate vs. visual distance	$r = -0.33$; LSO Δ MAE $+0.008$ ($p=0.053$), marginal	SA-FARI

4.11 Cross-model replication: the model swap

The two datasets above both vary the *data* while holding the tracker fixed at SAM3, so they leave one threat open: the null might be a quirk of SAM3 rather than a property of the prediction problem. A *controlled model swap* closes it — hold the cells, the ground truth, the four distances and the GLM fixed, and change *only the tracker*. If the before-running distances fail to predict a *different* tracker’s transfer too, the null is task-general. We swap in two open-vocabulary, text-promptable trackers of deliberately different architecture: **GLEE** [22], a single unified detect–segment–track model, and **Florence-2** [23] + SAM2 [17], a two-stage pipeline in which a caption-style generator emits boxes (with *no* confidence score) and SAM2 turns them into tracked masks.

SA-FARI cannot host the swap — and it is not the scorer. On SA-FARI both challengers score $p\text{DetA} \approx 0$ and cannot be modelled at all. This is not a rigged evaluator: the zeros survive box-HOTA (scoring the box models on boxes, GLEE 0.033) and a threshold-free average-precision metric with no confidence gate at all (GLEE 0.000, Florence-2 0.017–0.022); Florence-2, which emits no score, is unaffected by the gate and still scores ~ 0 . The cause is a *double confound*. SA-FARI is SAM3’s own co-released benchmark, and its exotic, nocturnal species are out-of-distribution for internet-trained challengers — and the domain, not merely

the species, is what breaks them: run GLEE on SA-FARI’s *everyday* species (jaguar, cow, fox) and it localises them well (oracle IoU up to 0.90) yet its confidence still collapses below any usable operating point on the dark, cluttered camera-trap footage. SA-FARI therefore cannot yield a fair, non-degenerate challenger score.

BURST hosts a fair swap, and both challengers work. BURST is neither SAM3’s benchmark nor camera-trap footage, and SAM3 scores *higher* on it (pDetA 0.607) than on its own SA-FARI (0.537) — so there is no home-turf inflation, and the arena is fair. There, both challengers genuinely track: GLEE reaches pDetA 0.340 (median 0.31, spread 0–0.98, 67/132 cells > 0.3) and Florence-2 + SAM 2 reaches 0.394 (median 0.33, 72/132 cells > 0.3), both non-degenerate with the real spread the regression needs (Table 4.16).

The null replicates across models. Under the same leakage-free leave-species-out cross-validation, size-controlled, the combined four-distance model fails to predict *either* challenger’s detection transfer: +0.011 (95% CI [−0.004, +0.027], $p = 0.088$) for GLEE and +0.034 ([−0.001, +0.068], $p = 0.030$) for Florence-2 — both confidence intervals spanning zero, neither clearing the Bonferroni bar ($\alpha = 0.0125$). As on SA-FARI, a few individual distances flicker (for Florence-2, environment $p = 0.006$ and temporal $p = 0.015$), but the discounts are identical: BURST has no locations, so “leave-species-out” is the only valid scheme and an apparent two-scheme pass is one partition counted twice; the flickers route through the animal-size confound the driver controls; and the *combined* model — what a deployment gate would use — is null in both. Two architecturally distinct trackers thus both (i) genuinely track BURST animals and (ii) show the before-running distance null. **The null is task-general across trackers, not a SAM-3 artefact.**

Table 4.16: Cross-model swap on BURST (132 cells, 41 species, mask-HOTA). *Top:* zero-shot pDetA per tracker — SAM3 and two architecturally distinct challengers, all non-degenerate. *Bottom:* out-of-sample error of the combined four-distance model (leave-species-out, size-controlled) vs a mean predictor; ΔMAE [95% CI] (p), Bonferroni $\alpha = 0.0125$. Both challengers track BURST animals yet the distance model is null for each — the null replicates across trackers.

<i>Zero-shot detection on BURST</i>			
Tracker	pDetA (mean / median / cells >0.3)		
SAM 3 (reference)	0.607		
GLEE (unified query)	0.340 / 0.31 / 67 of 132		
Florence-2 + SAM 2 (caption)	0.394 / 0.33 / 72 of 132		
<i>Combined four-distance model $\rightarrow$ pDetA (leave-species-out)</i>			
Challenger	ΔMAE	95% CI	p
GLEE	+0.011	[−0.004, +0.027]	0.088
Florence-2 + SAM 2	+0.034	[−0.001, +0.068]	0.030

Convergent, honestly bounded. This is a *controlled model swap*, not an independent replication: it reuses BURST’s cells, ground truth, distances and GLM, changing only the tracker. Its scope is two models on one fair dataset ($n = 132$, underpowered for a small species-level effect, exactly as the SAM3 BURST run), and Florence-2’s combined interval sits closer to zero-exclusion than GLEE’s — so we read it as a *convergent, confound-free* cross-model replication that removes the “single frozen backbone” threat, not as an independently-powered certification. Combined with the two datasets, the null now holds across both axes that could have confounded it — the dataset and the model.

Chapter 5

Discussion

5.1 Interpretation

Neither hypothesis is supported. H1 (detection falls with species novelty) and H2 (association falls with environment difficulty) both fail the pre-registered bar: no label-free distance beats a mean predictor out of sample, and the one coefficient that reached significance — visual distance, running in the opposite direction to the hypothesis (more-novel species detected better, not worse) — is an animal-size artefact. The follow-up intrinsic-difficulty pivot fares no better: low-light, clutter and night/IR predict detection only by proxying animal size, which alone carries the out-of-sample gain. So the honest reading is H0: transfer is *not* distance-driven, and the only label-free property that predicts detection is object size — the very quantity introduced as a nuisance control.

The detection–association decomposition, the intellectual core of the design, has a second, quieter finding: association is a near-degenerate target on this data. Because most cells contain a single object, **pAssA** tracks **pDetA** almost exactly, leaving little identity-specific variance for any feature to predict. The decomposition is therefore honestly a detection story; we report the association null with those divergence figures as its explanation rather than as a separate failed search.

5.1.1 Novelty at the extremes versus graded prediction

A natural objection is that novelty *surely* matters — a familiar species is detected better than a totally alien one, so how can a novelty distance be null? Both statements are true and do not conflict, because they concern different resolutions of the same axis. The *coarse* claim — that the extreme tails differ, alien-versus-familiar — is real and never in dispute; the null we report is the far stronger *graded* claim H1 actually tests: that a continuous distance predicts *where on the pDetA scale* an unseen species lands, out of sample, beyond the size confound, well enough to serve as a deployment gate. These come apart because the informative variance in **pDetA** lives in the middle of the range, not at the poles: the easy extremes (common large animals near 1, hopeless cases near 0) are already predicted by the mean, so a coarse “alien-versus-familiar” signal adds nothing exactly where prediction has value. A genuine extreme-tail effect can therefore coexist with a null graded predictor without contradiction — and it is the graded, held-out, size-controlled predictor, not the truism, that a practitioner would need and that this work shows does not exist. (The model swap sharpens this: the challengers’ own detect-or-fail behaviour on exotic species is itself a coarse novelty effect, yet it is *that model’s* novelty, measured against the wrong reference, and it too fails to yield a graded predictor — so it re-derives the hypothesis, not its confirmation.)

5.2 Relation to prior label-free evaluation

The null sits against a literature of *positive* results, and two axes separate every one of those results from ours — which is why the disagreement is only apparent. The first axis is the **signal**. Accuracy- and Agreement-on-the-Line [2, 14], ATC [8] and AutoEval [7] all read the model’s *own behaviour on the target* — its confidence, its agreement across a model pair, or (for AutoEval) a feature-distribution distance computed after passing the target images through the network. Those are *after-running* estimators. We instead forecast from *external*, label-free distances known *before* a single frame is processed — the strictly harder, and for a practitioner deciding whether to spend compute, the more useful regime. The second axis is the **output**. Every predecessor targets a *scalar* classifier accuracy [2, 7, 8, 14] or the quality of a *static* detector/segmenter [20, 25]; none concerns identity maintained over time, and none is decomposed. Ours is the first study to ask the question of a promptable *video tracker*, split into detection (pDetA) and association (pAssA). AutoEval [7] is the closest methodological cousin — it, too, regresses a performance score on a distributional feature distance — and the strength of its positive sharpens the contrast: it reports a near-perfect rank correlation (Spearman $\rho \approx -0.91$) between a Fréchet distribution-shift distance and classifier accuracy. That distance, though, is computed from the model’s own penultimate features *after* the target images are passed through the network, and predicts a scalar classifier accuracy. We test AutoEval’s exact functional in our own regime — the Fréchet and MMD distributional distances on cached DINOv2 embeddings, computed *before* running SAM3 (§4.9, Table 4.11) — and it stays null ($\Delta\text{MAE} \leq 0.007$, $p \geq 0.10$, every confidence interval spanning zero), so the failure is not an artefact of a lazy nearest-prototype distance: even the strongest AutoEval-style distance carries no before-running transfer signal. The contrast is not merely one of magnitude but of *sign*: our nearest analogue, the visual distance, correlates with detection in the *opposite* (wrong) direction and dissolves under an animal-size control. Our result therefore does not contradict AutoEval’s positive; it extends the recipe to a regime — distance measured before running, target a decomposed video tracker — where the shortcut stops working, and reports honestly that it does. A final bridge makes the boundary concrete: Accuracy-on-the-Line [14] was validated on iWildCam, the same camera-trap shift this dissertation cites via WILDS and Terra Incognita [3, 11] — so the exact wildlife shift where a classifier’s OOD accuracy *is* predictable label-free is where our before-running distances are *not*, across both the classifier-to-tracker and after-running-to-before-running boundaries.

5.3 Reconciling the environment null with the background-bias literature

Our environment axis inherits its motivation and its foreground/background separation from a literature that finds backgrounds emphatically *do* matter [3, 4, 18, 24], so the null it returns deserves an explicit reconciliation rather than a bare report. The apparent tension dissolves on two distinctions. The first is **causal reliance versus predictive distance**. Xiao et al. [24], Rosenfeld et al. [18] and PanAf-FGBG [4] all *intervene* on the background — swap it, transplant the object, or subtract the paired background-only clip’s features — and watch performance move; PanAf, most sharply, shows a chimpanzee-behaviour classifier that scores from the background *alone* almost as well as from the foreground on unseen locations, and recovers several points of the location drop by subtracting the background. That is a statement about how a model *uses* context. Ours is the strictly weaker, correlational question of whether a scalar scene *distance*-from-reference *forecasts* an untouched model’s per-cell score — and a model can lean causally on context while its score stays unpredictable from a one-number distance. Our null therefore does not deny their causal findings; it locates a regime where the label-free-distance shortcut, not the context reliance, fails.

The second distinction is **why the promptable regime should be the one where it fails**, and here the same literature supplies the mechanism. Xiao et al. [24], Rosenfeld et al. [18] and PanAf all study models that must *infer* the category — a classifier with no foreground cue, or a detector reading co-occurrence — for which a spuriously correlated background is a usable shortcut. SAM3 is *told* what to find: the prompt "impala" supplies the foreground identity, removing the incentive to ride the scene, exactly the direction Xiao et al. [24]’s own finding predicts (better, less-shortcut-prone models close the background gap). PanAf’s target compounds this — fine-grained *behaviour* is intrinsically background-correlated (a chimp nests in trees, drinks at water), whereas ours is finding and following the *named animal itself*, where naming it short-circuits the context dependence. The honest reading is thus an extension, not a refutation: the very backgrounds that causally drive a camera-trap classifier’s location drop do not, once surgically isolated from the animal and used as a before-running distance, forecast a promptable tracker’s transfer — and because we isolate the scene from the animal explicitly (a step none of the label-free-prediction methods take), this is a cleaner statement than “background is diagnostic”, not a weaker one. The claim is bounded accordingly: it concerns background *distance* as a *predictor* of *detection* transfer for a promptable tracker (the secondary location split, replicated null on BURST, with a near-degenerate association target), not a blanket “background is irrelevant”, which would wrongly contradict [3, 4, 24].

5.4 Why label-free distances do not yield a deployment estimator

The field problem that motivated this work — telling a practitioner, before spending compute, whether SAM3 is trustworthy for a new (species, place) — is not solved by label-free distances. That is a negative but useful answer: the intuition that “far-from-training implies unreliable” does not hold for a promptable video tracker, so a deployment gate cannot be built from taxonomic, visual, environmental or temporal distance to a reference set alone. The one predictive label-free signal, animal size, is too coarse and too obvious to serve as such a gate. What the negative result does establish is a rigorous baseline: any future before-running estimator must clear the same whole-group hold-out bar that these well-motivated distances failed.

5.5 Limitations and threats to validity

Three limitations bound the claim. First, distances are measured against the SA-FARI train split, *not* SAM3’s true pretraining corpus (Meta’s SA-Co), which is undisclosed; the null is scoped to distance-from-SA-FARI. We test one hedge against this — the familiarity proxy (§4.8), which reads separability directly from SAM3’s own features — and it too is null, recovering the same size-entangled visual signal, so even model-internal familiarity does not lift the null; a coverage-anchored proxy for the actual corpus nonetheless remains untested. Second, the analysis is a single dataset and a single frozen backbone, and — since one inference pass is reused across both splits — the species and location experiments are two overlays on the *same* scored cells, not independent replications; they give near-orthogonal but not statistically independent evidence. We take two steps against this. Against the *dataset* threat we re-run the whole pipeline on an independent dataset: on the larger, mask-native BURST (§4.9, 132 cells / 41 species) the distance null holds again, giving a second convergent null (not positive proof — BURST detects only a large between-species predictor and remains underpowered for a small species-constant distance effect — so we report convergence, not independent confirmation of any positive). Against the *model* threat — the “single frozen backbone” — we run the controlled model swap (§4.11): two architecturally distinct open-vocabulary trackers, GLEE and Florence-2 + SAM2, both track BURST’s animals non-degenerately (pDetA 0.34 and 0.39) yet the combined distance

model is null for each (CIs span zero, neither clears the Bonferroni bar), so the null is task-general rather than a SAM 3 artefact — a convergent, confound-free cross-model replication (again not an independently-powered certification: two models on one fair dataset). That swap could not run on SA-FARI itself, and honestly so: SA-FARI is SAM 3’s own co-released benchmark, and its camera-trap domain collapses the challengers’ confidence calibration even where they localise well, so BURST is the fair-swap ceiling. Third, the effective sample is tens of independent species/locations, not hundreds of cells — which is why inference is group-clustered throughout, and why aggregate correlations (e.g. the location-level night/IR effect) are reported alongside their much weaker cell-level values to avoid the ecological-correlation fallacy. Representational probing is only partly opened here: the familiarity proxy (§4.8) probes whether SAM 3’s features *separate* the species — they do, but that separability merely re-encodes the visual distance and does not predict transfer — while a fuller probe of what those representations encode remains open.

Finally, on the scorer itself: all scores come from the *unmodified official* SA-FARI evaluator, which wraps the standard HOTA/TETA tracking metrics rather than any SAM-3-specific logic. Its one SAM-3-calibrated knob is a fixed 0.5 confidence gate; we verified it is not load-bearing for the cross-model comparison, since the challengers’ near-zero SA-FARI scores survive both a box-based metric and a threshold-free average-precision metric with no gate at all (§4.11). The measurement is therefore model-fair, and the challenger failures on SA-FARI are genuine detection failures, not scoring artefacts.

5.6 Ethical considerations

The data are anonymised camera-trap footage released under a non-commercial licence; location identifiers are already anonymised in SA-FARI. The negative result carries a responsible-deployment message: because label-free distance does *not* reliably flag when SAM 3 will fail, conservation teams should not treat low-distance-from-training as a safety guarantee, and reliability should be checked with labelled spot-audits rather than assumed.

Chapter 6

Conclusion

6.1 Summary of findings

We asked whether SAM 3’s zero-shot promptable-tracking accuracy on an unseen (species, place) can be predicted, before running the model and without target labels, from four label-free distances — separately for detection and association. Under a deliberately powerful out-of-sample test the answer is no: no distance beats a mean-predictor baseline on either a species or a location hold-out, the one apparent effect is an animal-size confound, and a follow-up intrinsic-difficulty pivot collapses to the same size effect. Only object size predicts detection, modestly, and it is the nuisance the analysis controls for. Association carries almost no signal distinct from detection. The null is robust to the two ways it might have been an artefact: it holds across two datasets (SA-FARI and BURST), and, in a controlled model swap on BURST, across two architecturally distinct promptable trackers, so it reflects the prediction problem rather than any one dataset or model. The result is a rigorous, decomposed, and now cross-model null.

6.2 Contributions

To our knowledge this is the first study to test, out of sample, whether a *promptable video tracker’s* zero-shot transfer is predictable from label-free, before-running signals, and the first to decompose that question into detection and association. Its contribution is the honest negative answer and its explanation: transfer is not distance-driven; the apparent signals reduce to an animal-size confound; the test is shown to detect that one real signal when present, so the null reflects an absence of signal, not of power; and the null generalises — across two datasets and, via the controlled model swap, across two other trackers — so it is task-general rather than specific to SAM 3.

6.3 Future work

Three directions remain open. First, a fuller *representational* probe. The familiarity proxy already tests whether SAM 3’s own features *separate* the species and finds they do, but only by re-encoding the visual distance (Section 4.8); what those features actually *encode* — whether taxonomy is represented structurally, and a coverage-anchored proxy for the true, undisclosed pretraining corpus — is left untested. Second, broadening the model swap: more trackers, and ideally a *larger* fair, mask-native dataset than BURST that is not any tracker’s home turf — SA-FARI’s larger cell count could not be reused, because its camera-trap domain collapses internet-trained challengers’ calibration even where they localise well, making it the fair-swap ceiling here. Third, targets with richer association structure (crowded, multi-object scenes) where the detection–association decomposition would have more to distinguish.

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Appendix A

Mathematical definitions

This appendix collects the exact definitions used in Chapter 3. The tracking metric is computed by the official evaluator and never re-implemented; the distances and the statistical procedures are ours and are stated here as they appear in the code.

A.1 The promptable-tracking metric

Promptable HOTA decomposes, at each localisation threshold α , into a detection and an association term [12]:

$$\text{HOTA}_\alpha = \sqrt{\text{DetA}_\alpha \cdot \text{AssA}_\alpha}, \quad \text{pHOTA} = \frac{1}{|\mathcal{A}|} \sum_{\alpha \in \mathcal{A}} \text{HOTA}_\alpha, \quad (\text{A.1})$$

with $\mathcal{A} = \{0.05, 0.10, \dots, 0.95\}$. We analyse the two components separately as the targets $\text{pDetA} = \text{DetA}$ (did it *find* the animals?) and $\text{pAssA} = \text{AssA}$ (did it *keep* each identity?), both bounded in $[0, 1]$. Scores are computed per cell by the vendored **VEval** evaluator.

A.2 Label-free distances

Let the frozen reference be the “seen” set of a split; every distance below is computed against that reference only.

Taxonomic distance. Write each species’ seven-level path $\pi(s) = (\text{kingdom}, \dots, \text{species})$, defined only for species with a complete taxonomy. Let $\text{shared}(a, b)$ be the length of the longest common *non-empty* prefix of two paths. The tree-step distance and the nearest-reference distance are

$$\tau(a, b) = L - \text{shared}(a, b) \quad (L = 7), \quad d_{\text{tax}}(s) = \min_{s' \in \mathcal{R}, s' \neq s} \tau(\pi(s), \pi(s')), \quad (\text{A.2})$$

where the exclusion $s' \neq s$ is the leave-one-species-out rule (Split A). Shared genus gives $\tau = 1$, family 2, order 3, fully disjoint 7; species without a full taxonomy take $d_{\text{tax}} = \text{NaN}$.

Prototype and nearest cosine distance. Given a set of ℓ_2 -normalised embeddings $\{v_i\}_{i=1}^N$, the prototype and the distance from a query v to a set of prototypes $\{p_j\}$ (optionally excluding an index e) are

$$p = \frac{\bar{v}}{\max(\|\bar{v}\|_2, 10^{-12})}, \quad \bar{v} = \frac{1}{N} \sum_i v_i, \quad d(v) = 1 - \max_{j \neq e} v^\top p_j. \quad (\text{A.3})$$

Because v and each p_j are unit-norm, $v^\top p_j$ is the cosine similarity, so $d \in [0, 2]$.

Visual and environment distance. Both use DINOv2 embeddings. The *visual* distance embeds the ground-truth-mask-cropped animal (the background is zeroed before cropping), forms one prototype per species, and sets $d_{\text{vis}}(s)$ to the nearest *other* species’ prototype. The *environment* distance embeds the animal-masked-out background (the animal union is replaced by the mean colour of the remaining pixels), forms one prototype per location, and sets $d_{\text{env}}(\ell)$ to the nearest other location. Two interpretable scene covariates accompany the environment distance:

$$\text{achromatic}(f) = \left[\overline{\max_c f_c - \min_c f_c} < \tau_{\text{ir}} \right], \quad \tau_{\text{ir}} = 12, \quad (\text{A.4})$$

$$\text{is_night_ir}(\ell) = \left[\overline{\text{achromatic}(f)} > 0.5 \right], \quad \text{clutter}(\ell) = \overline{\text{Var}(\nabla^2 f)}, \quad (\text{A.5})$$

where the bar denotes a mean over a location’s frames, $\max_c - \min_c$ is the per-pixel channel spread, and ∇^2 is the four-neighbour Laplacian on the greyscale background.

Temporal gap and animal size. For a cell with footage year y and reference years $\mathcal{Y}$, and for the size covariate,

$$d_{\text{temp}} = \min_{r \in \mathcal{Y}} |y - r|, \quad \log_area(s) = \ln(1 + \overline{\text{area}(s)}), \quad (\text{A.6})$$

where $\overline{\text{area}(s)}$ is the mean ground-truth mask pixel-count over sampled masklets of species s . Size is a *nuisance* covariate, not a hypothesis variable.

A.3 Model and statistics

Support-weighted fractional-logit GLM. Each bounded target $y \in [0, 1]$ is modelled with a Binomial (logit-link) GLM

$$\text{logit } \mathbb{E}[y] = x^\top \beta, \quad \text{fit with variance weights } w = n_{\text{frames}}, \quad (\text{A.7})$$

where the design x standardises each continuous predictor $z = (x - \mu)/\sigma$ (missing values imputed to the training mean μ , zero-variance columns dropped), adds $\log_{\text{support}}$, the standardised $\ln(1 + n_{\text{frames}})$, passes binary `is_night_ir` through, and adds an intercept. Standardisation statistics are learned on training rows only.

Explained deviance and the cross-validation gain. Fit quality is the McFadden pseudo- R^2 , and per-cell out-of-sample gain over a mean predictor is

$$R_{\text{McF}}^2 = 1 - \frac{D}{D_0}, \quad \delta_i = \underbrace{|y_i - \bar{y}|}_{\text{baseline error}} - \underbrace{|y_i - \hat{y}_i|}_{\text{model error}}, \quad (\text{A.8})$$

with D, D_0 the model and null (intercept-only) deviances; $\delta_i > 0$ means the distance model beats “guess the mean” on cell i .

Weighted Pearson correlation. For the species-level false-positive analysis, with weights w (per-species probe counts),

$$r_w = \frac{\sum_i w_i (x_i - \bar{x}_w)(y_i - \bar{y}_w)}{\sqrt{\sum_i w_i (x_i - \bar{x}_w)^2} \sqrt{\sum_i w_i (y_i - \bar{y}_w)^2}}, \quad \bar{x}_w = \frac{\sum_i w_i x_i}{\sum_i w_i}. \quad (\text{A.9})$$

Appendix B

Algorithms

Pseudocode for the four non-obvious procedures. The two bootstraps differ deliberately: the cross-validation gain (Alg. 3) is tested *one-sided* (does the model beat the mean?), whereas the false-positive correlation (Alg. 4) is tested *two-sided* (does hallucination change with novelty, in either direction?).

Algorithm 1: Memory-lean, cache-first crop embedding (`embed_crops`).

Input: items (key, record, frame, crop_fn); embedding cache; embedder; chunk size C , workers W

Output: $\{\text{key} \mapsto \text{embedding}\}$

```
1 misses ← [item : cache.get(item.key) = ∅];
2 for each contiguous chunk  $B$  of  $C$  items in misses do
3   group  $B$  by video; for each video do ensure_frames(missing frames);
4   loaded ← map load_frame → crop_fn over  $B$  using  $W$  threads;
5   pending ← [( $k, c$ ) ∈ loaded :  $c \neq \emptyset$ ];
6   if pending ≠ ∅ then
7     vecs ← embedder.embed(crops of pending);
8     for ( $k, \_$ ),  $v$  in pending, vecs do cache.put( $k, v$ );
9   free pending ;                               // bound peak RAM to one chunk
10  every 16 chunks: cache.save() ;                // resumable checkpoint
11 if misses ≠ ∅ then cache.save();
12 return {item.key ↦ cache.get(item.key)} for all items;
```

Algorithm 2: Group-cluster-bootstrap coefficient CIs (`group_bootstrap_cis`).

Input: scored cells; target; grouping columns $\mathcal{G} = (\text{category_id}, \text{location_id})$; resamples B ; level $\alpha = 0.05$

Output: per-coefficient interval $[\text{ci_lo}, \text{ci_hi}]$

```
1 foreach grouping column  $g \in \mathcal{G}$  present in the data do
2    $U \leftarrow$  unique groups of  $g$ ; if  $|U| < 2$  then skip;
3   for  $b \leftarrow 1$  to  $B$  do
4     draw  $|U|$  groups from  $U$  with replacement; sample  $\leftarrow$  all rows of the drawn
       groups;
5     refit the GLM on sample; record its coefficient vector ;    // skip a singular
       resample
6    $\log \leftarrow$  percentile  $\frac{\alpha}{2}$ ,  $\text{hi}_g \leftarrow$  percentile  $1 - \frac{\alpha}{2}$ , per coefficient;
7 return conservative envelope  $\text{ci\_lo} = \min_g \log$ ,  $\text{ci\_hi} = \max_g \text{hi}_g$ ;
```

Algorithm 3: Paired group-bootstrap cross-validation significance (`oos_predictions` + `_bootstrap_delta`).

Input: scored cells; target; group column (species or location); resamples B

Output: ΔMAE , its 95% CI, and one-sided p

/ Leave-one-group-out out-of-sample prediction*

**/*

```
1 foreach held-out group do
2   refit the GLM on the other groups (own standardisation); predict the held-out cells ;
   // assert train/test groups disjoint
3  $\delta_i \leftarrow |y_i - \bar{y}| - |y_i - \hat{y}_i|$  for every predicted cell;
   /* Group bootstrap of the mean gain
4 for  $b \leftarrow 1$  to  $B$  do
5   draw whole groups with replacement;  $\text{boot}_b \leftarrow \text{mean}(\delta \text{ over the drawn cells})$ ;
6  $\text{CI} \leftarrow$  percentiles  $[2.5, 97.5]$  of  $\text{boot}$ ;  $p \leftarrow \Pr(\text{boot} \leq 0)$ ; significant  $\Leftrightarrow \text{CI\_lo} > 0$ ;
7 return  $\Delta\text{MAE} = \text{mean}(\delta)$ , CI,  $p$ ;
```

Algorithm 4: False-positive / hallucination analysis (`false_positives`).

Input: split; score threshold t ; resamples B

Output: overall FP rate; per-species FP rate vs. novelty with a bootstrap CI

```
1  $\mathcal{H} \leftarrow$  hard-negative probes (queried species absent);
2 build a partition whose probe species are the hard-negative species; compute  $d_{\text{tax}}, d_{\text{vis}}$ 
   for them;
3 foreach probe (video, species)  $\in \mathcal{H}$  with a prediction file do
4    $\text{fp} \leftarrow [\max_k \text{score}_k \geq t]$ ;    // a masklet returned for an absent species = a
       hallucination
5   attach  $d_{\text{tax}}, d_{\text{vis}}$  of the species;
6 aggregate to per-species FP rate;  $r \leftarrow \text{weighted-Pearson}(\text{FP rate}, \text{distance})$ ;
7 for  $b \leftarrow 1$  to  $B$  do
8   resample species with replacement; recompute  $r$  ;
9  $\text{CI} \leftarrow$  percentiles  $[2.5, 97.5]$ ; significant  $\Leftrightarrow \text{CI}$  excludes 0 (two-sided);
10 return overall FP rate, taxonomic-tercile breakdown,  $r$  and CI for taxonomic and
    visual novelty;
```

Appendix C

Supplementary results